

Treatment of Human African Trypanosomiasis in 2019

Translational Considerations from Clinical Trials to Clinical Practice

1. Introduction

Treatment of Human African Trypanosomiasis (HAT) has improved dramatically over the last two decades. The field has moved from toxic and less effective therapies to less toxic highly effective therapies. An ongoing concern is the cost associated with distribution and administration (including the need for a staging lumbar puncture) and other logistical issues associated with current parenteral therapies such as Nifurtimox-Eflornithine Combination Therapy (NECT) and pentamidine.

Fexinidazole (FEXI) is a new chemical entity for the oral treatment of HAT. Fexinidazole is a nitroimidazole — the same class to which metronidazole and tinidazole belong. Fexinidazole is orally bioavailable and readily metabolized to two metabolites. Both metabolites and the parent compound are bioactive against *Trypanosoma brucei gambiense*. The pharmacokinetic profile is sufficient to permit once daily dosing in the presence of food over a period of 10 days.

A single pivotal Phase 2/3 study (FEX004) in late Stage 2 HAT patients was conducted and served as the basis for submission of a dossier to the European Medicines Agency for review under Article 58 of Regulation 726/2004. A positive scientific opinion was adopted by the Committee for Medicinal Products for Human Use (CHMP) on 15 November 2018.¹ Subsequently marketing authorization of fexinidazole was granted in the Democratic Republic of Congo (DRC) on January 30, 2019.²

In addition to the pivotal study, a study in earlier Stage patients (FEX005) and a study in pediatric subjects (FEX006), as well as an ongoing Phase 3b study in subjects with Stage 1 and 2 HAT (FEX009), provide supportive data for safety and effectiveness in Stage 1 and Stage 2 HAT. In all, the safety and effectiveness of fexinidazole were evaluated in three clinical studies involving 749 subjects across the different stages of disease.

The pivotal study, conducted by Drugs for Neglected Diseases initiative (DNDi), was a randomized, positive-controlled, non-inferiority, open-label study comparing FEXI with NECT at an enrollment ratio of 2:1. The trial demonstrated that FEXI is effective at curing the disease with success defined as no evidence of trypanosomes in any body fluid, not requiring rescue medication, and having a cerebrospinal fluid white blood cell count ≤ 20 cell/ μ L at 18 months. Statistical non-inferiority to NECT was achieved.

¹ CHMP recommends first oral-only treatment for sleeping sickness | European Medicines Agency. Downloaded 10 April 2019

² <https://www.globenewswire.com/news-release/2019/01/30/1707678/0/en/Sanofi-Fexinidazole-the-first-all-oral-treatment-for-sleeping-sickness-approved-in-Democratic-Republic-of-Congo.html>. Downloaded 20 March 2019

The full approved indication is:

Fexinidazole Winthrop is indicated for the treatment of both first-stage (haemo-lymphatic) and second-stage (meningo-encephalitic) of human African trypanosomiasis (HAT) due to Trypanosoma brucei gambiense in adults and children ≥ 6 years old and weighing ≥ 20 kg.³

Fexinidazole represents a significant advance in the stage-independent treatment of HAT, in essence obviating or drastically reducing the need for a staging lumbar puncture and parenteral administration of medications for Stage 1 HAT (Pentamidine IM or IV) or Stage 2 HAT (IV eflornithine with oral nifurtimox).

Policy considerations:

As noted, in the CHMP communication EMA/791484/2018 of 15 November 2018 on the basis of the FEX004 results only (vs. totality of data from prior studies in HAT with NECT), the summary of opinion listed the following caveat: *“Patients with more severe central nervous system involvement are deemed at higher risk of failing treatment with fexinidazole.”⁴*

Furthermore, in the Press release EMA/CHMP/798734/2018 dated 16 November 2018, it was noted in patients *“whose central nervous system is already severely affected, Fexinidazole should only be given under strict supervision in hospital when no other adequate treatment is available or tolerated.”*

Finally, in the same release: *“Because there are a small number of cases of late relapse in the studies, the CHMP recommends a follow-up monitoring of up to 24 months to ensure the surveillance of potential relapses.”*

The above statements are in response to the numerical increase in treatment failures in the FEXI arm of study FEX004 in the “more severely” ill HAT patients. These statements require further inquiry when relating clinical trial methodology and results into policy and ultimately clinical practice for the use of Fexinidazole. These further inquiries should include the totality of the data for various options for the two most efficacious options – FEXI and NECT.

Additionally, subsequent questions to inform policy and clinical practice that include the following should be addressed:

- How does severity of disease influence treatment success?
- What is the time course and percentage of patients with improvement of symptoms and biologic markers of infections?
- What constitutes more severe central nervous system involvement and can the patients with more severe disease be identified through non-invasive means (e.g. a clinical symptom or symptom score with reasonable sensitivity and specificity comparable to LP results)?
- What is the definition of relapse?

³ https://www.ema.europa.eu/documents/smop-initial/chmp-summary-opinion-fexinidazole-winthrop_en.pdf.
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⁴ https://www.ema.europa.eu/documents/smop-initial/chmp-summary-opinion-fexinidazole-winthrop_en.pdf

- What is the time course over which relapse is determined, and how does relapse look clinically based on the clinical trial definition (e.g., rapid and severe vs indolent and mild)?
- How does the time to relapse compare between treatments?
- What is the time course of CSF WBC change following treatment?
- What is the time course of parasite clearance, especially those parasites detected in the CSF?

Finally, the Assessment Report on Fexinidazole Winthrop (EPAR; EMA/CHMP/843546/2018;) published by the Committee for Medicinal Products for Human Use (CHMP) on 15 November 2018 focused heavily, if not exclusively, on outcomes at 18 months post-treatment. The protocol-defined secondary efficacy endpoints of outcome of treatment as assessed on Day 11 (24 hours after EOT), and 3, 6, 12, and 24 months, after the EOT were not reported; hence the clinician has no information to assess time-related outcomes.⁵ We have included these outcomes below.

2. Time course of effect

In FEX004, assessments of outcome of treatment (as a binary outcome of either success or failure) were done at Day 11 (24 hours after end of treatment; EOT) and 3, 6, 12, 18 and 24 months after EOT. The Failure Free Rates and 95% confidence intervals at each of these time points for FEXI and NECT are shown in the table below. FEX004 study shows that the cumulative failure rates over time are similar between the two treatments FEXI and NECT.

Treatment	Patients treated (N)	Failure-free count and rate at EOH [95% CI]	Failure-free count and rate at 6 months [95% CI]	Failure-free count and rate at 12 months [95% CI]	Failure-free count and rate at 18 months [95% CI]
Fexinidazole	262	260/262 99.2% [97.0; 99.8]	254/261 97.3% [94.5; 98.7]	249/261 95.4% [92.1; 97.4]	210/228 93.1% [89.3; 95.6]
NECT	127	127/127 100.0% [100.0; 100.0]	126/127 99.2% [94.5; 99.9]	124/126 98.4% [93.8; 99.6]	112/114 98.4% [93.8; 99.6]
P-value		0.9725	0.0700	0.0354	0.0410

Table 1. Data from DNDiFEX004 – fexinidazole – Clinical Study Report 28-Jul-2017 Version 8.0. (mITT) population. Rates and confidence intervals are based on the Kaplan-Meier analysis, while *p*-values are based on a repeated measures logistic regression model (data on file).

Clearly, both treatments are rapidly acting and provide a high rate of success by end of treatment. This figure would provide clinicians visual/graphical evidence consistent with the statistical finding of non-inferiority between the two treatments when considering the mITT population.

⁵ https://www.ema.europa.eu/en/documents/medicine-outside-eu/fexinidazole-winthrop-assessment-report_en-0.pdf Update 4/3/2019

Although statistical non-inferiority with a margin of 13% prospectively agreed to by the EMA was achieved in FEX004, there were proportionally more failures in the FEXI group compared to NECT at 18 months (8.8% vs 2.4%; N=264 vs 130; difference Fexinidazole–NECT: -6.42%; 97.06% CI [-11.22; -1.61]).⁶

Although sensitivity analyses indicated this difference was statistically significant, this difference is unlikely to be clinically or operationally significant, especially when viewed in context of the totality of NECT experience to date.

Looking beyond the FEX004 study, it is useful to consider the totality of information known about both Fexi and NECT. Specifically, how does the cumulative failure rate of fexinidazole and NECT in FEX004 compare with historical data from either NECT, melarsoprol or eflornithine monotherapy? Figure 1 (Figure 11 from DNDiFEX004 Clinical Study Report) provides insights through the Kaplan-Meier plot showing survival probability over time.

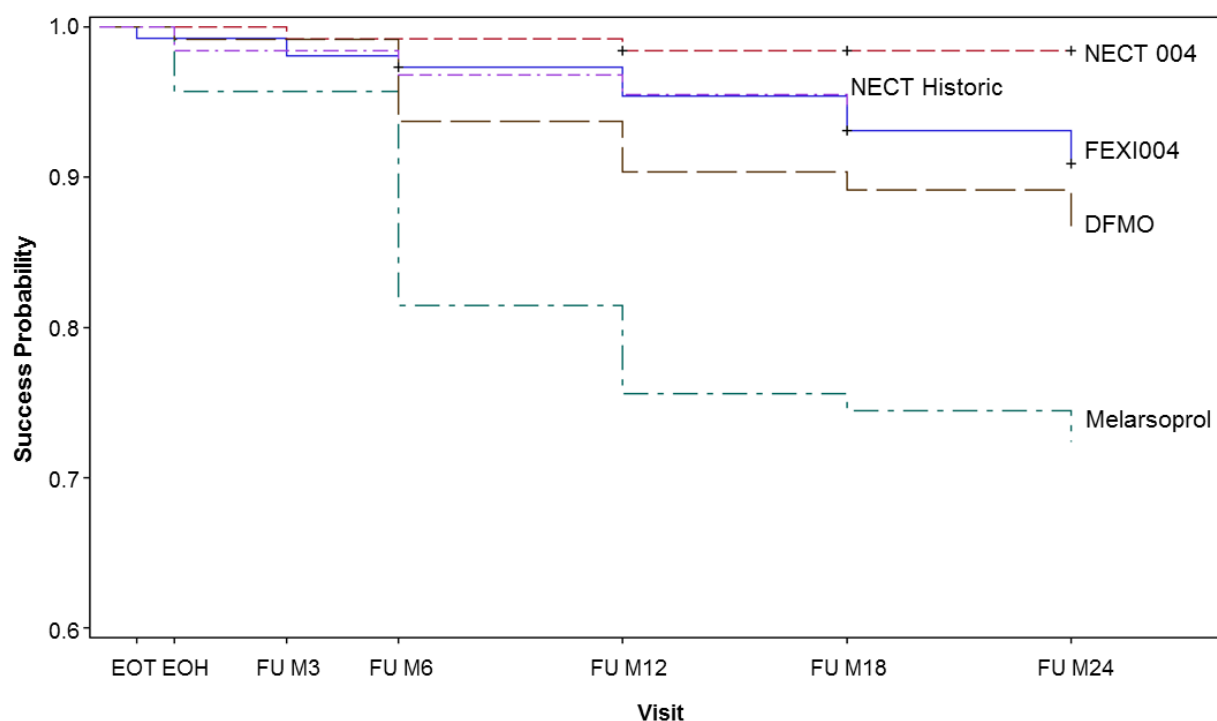


Figure 1: Kaplan-Meier Plot of Probability of Success. From Figure 11, DNDiFEX004 Clinical Study Report. Data on file.
DFMO = Eflornithine. NECT = Nifurtimox-Eflornithine Combination Treatment

The Failure Free Rates and 95% confidence intervals at Month 18 for the above Kaplan-Meier plot are also shown in the table below:

⁶ Fexinidazole Winthrop. Annex 1. Summary of Product Characteristics.

https://www.ema.europa.eu/en/documents/medicine-outside-eu/fexinidazole-winthrop-product-information_en.pdf

Cohort and Treatment	Patients Treated (N)	Month 18	
		Failure Free Rate	95% CI for Failure Free Rate
FEX004 NECT	127	98.43%	93.8% to 99.6%
Historic NECT	756	94.7%	92.9% to 96.2%
FEX004 Fexinidazole	262	93.11%	89.3% to 95.6%
Historic Melarsoprol	9043	74.45%	73.1% to 75.8%
Historic DFMO	1323	89.16%	87.0% to 91.3%

Table 2. Data extracted from Table 39, DNDiFEX004 Clinical Study Report. Data on file.

With regard to the Kaplan-Meier plot above (Figure 1), the survival curve for FEXI in FEX004 is superimposable on the curve for historical NECT. The survival curve for NECT in the FEX004 study was numerically better than FEXI, yet non-inferiority at the pre-specified margin was demonstrated overall between NECT and FEXI. Given the relatively small numbers of patients treated with NECT in FEX004 vs. the numbers in prior studies, the question arises as to how FEXI results compare to the totality of NECT experience to date. Since the clinical trial methodology and endpoints were similar across studies, this is a reasonable comparison to make.

When the rates of success and 95% confidence bands are plotted for fexinidazole (FEX004) and total NECT data (NECT in FEX004 combined with Historic NECT), the graph inserted below provides further perspective and context on the success rates over time of fexinidazole compared to the totality of NECT data. As with the Kaplan-Meier plots in Figure 11, the curve for FEXI in essence overlaps the NECT curve. The 95% confidence bands are wider for FEXI, reflecting the smaller sample size compared to total NECT; nevertheless the success rates of both NECT and FEXI are similar over time.

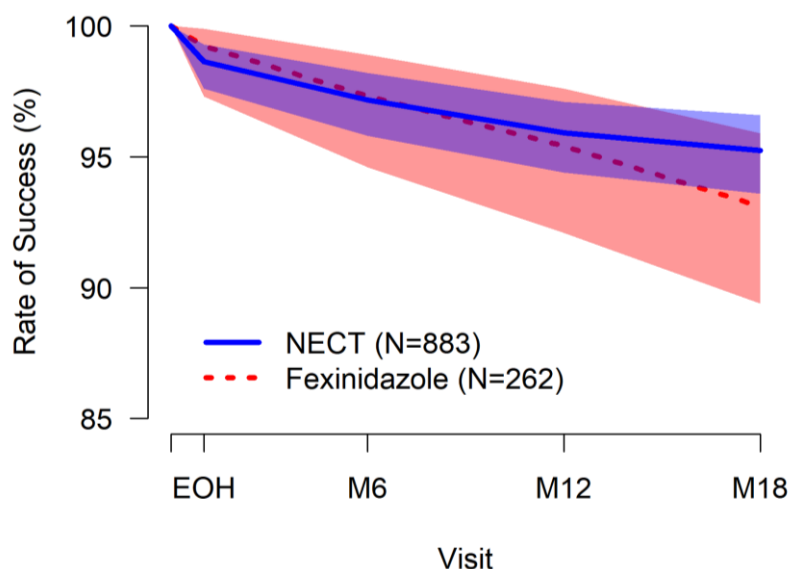


Figure 2. Rate of success ($\pm$ 95% CI) for fexinidazole in Study 004 and NECT in Study 004 combined with historical NECT data

The melarsoprol Kaplan-Meier curve in Figure 1 also provide a historical comparison and context. Whether the high failure rate for historical melarsoprol was due to intrinsic or extrinsic factors such as reported drug resistance is unknown. Nevertheless, with the relatively high risk:benefit ratio of melarsoprol, it is no wonder that staging of disease as Stage 1 vs Stage 2 with lumbar puncture was necessary so that patients could be adequately informed of the risk of death from reactive encephalopathy and also probability of relapse in a relatively short amount of time.

Lumbar puncture staging has continued since the advent of NECT since one treatment regimen is not used therapeutically for both stages of HAT.⁷ With the positive opinion for fexinidazole as an oral treatment shown to be effective for both Stage 1 and Stage 2, it could obviate the need for a lumbar puncture for therapeutic decision-making.

3. Time Course of Parasite Clearance

To assess the time course of parasite clearance, Trypanosome data from FEX-004, -005, -006 were plotted as follows:

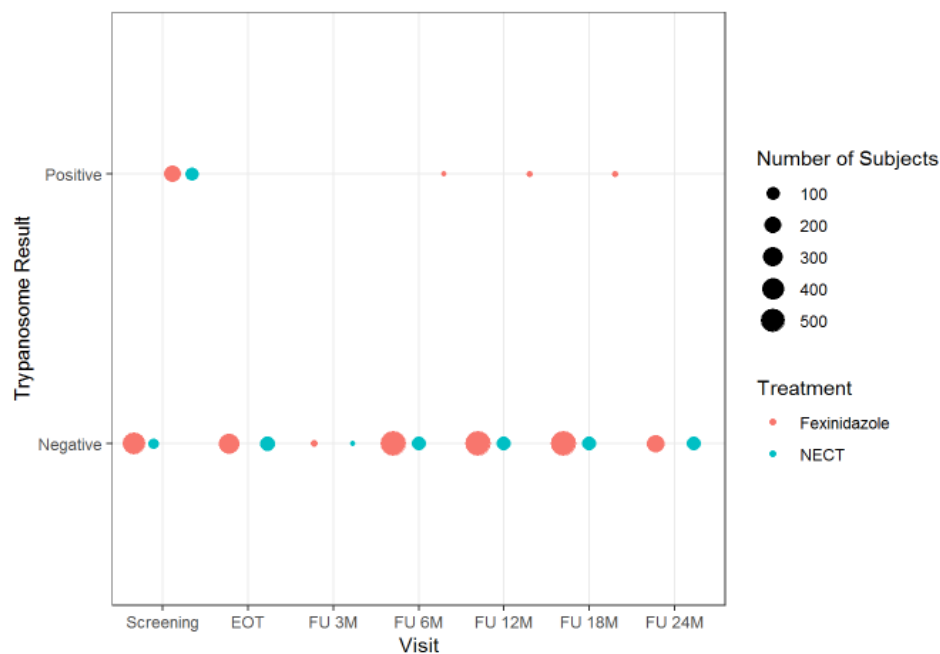


Figure 3: Time Course of Parasite Presence/Absence

The red dots reflect subjects who received fexinidazole, and the green dots those who received NECT. The size of the dot reflects the relative number of subjects in each category. The number of subjects positive or negative for trypanosomes at each visit are plotted

From this plot it is clear that parasite counts in the CSF became negative at end of treatment for both drugs in all subjects. For fexinidazole, one subject becomes positive at month 6, one more at month 12, and one at month 18. None of the NECT patients reverted to positive following treatment. Whether this is a function of drug effect or smaller NECT study population size is uncertain.

⁷ No apparent evidence in the literature that NECT is effective against Stage 1 HAT.

Overall, however, this graph provides assurance to clinicians about the rapid and persistent clearance of parasites over time for most (3/262) patients treated with fexinidazole.

The microscopic presence of parasites in any body fluid following treatment could either reflect recrudescence or reinfection. Without the availability of a validated technique to genotype parasites before and after treatment, the question of recrudescence or reinfection is uncertain.

4. Time Course of Change in CSF WBC

CSF leukocytic pleocytosis is a clinical laboratory finding in Stage 2 HAT. As a biomarker of severity of disease, it is semi-quantitative and has been the subject of much debate. Nevertheless, it is generally held by key opinion leaders that CSF WBC > 20 cells/ μ L is consistent with late Stage 2 HAT, whereas CSF WBC < 5 cell/ μ L is not consistent with Stage 2 HAT.

Figure 4 depicts the evolution of CSF WBC over time for the ITT population of FEX004. From the median plots $\pm$ Q1 and Q3, it is clear that there is substantial variability at baseline. However, by the end of treatment, median CSF WBCs have decreased and by month 6, 12, and 18, remain remarkably low among the treatment successes for both drugs. Whether the difference at EOT is clinically relevant is a subject of debate.

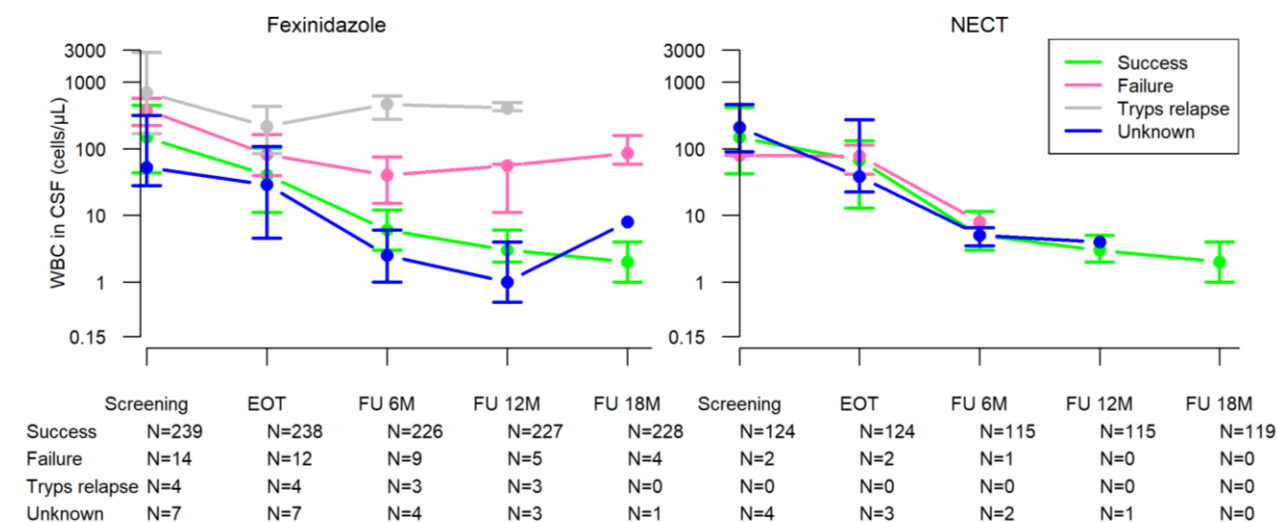


Figure 4. Median ($\pm$ Interquartile Range) CSF WBC count over time (ITT population). Data on file.

The reduction in median CSF WBC over time corresponded with improvements in prospectively obtained clinical signs and symptoms of HAT post-treatment over time. The following radar plots demonstrates improvement in signs and symptoms for both fexinidazole and NECT.

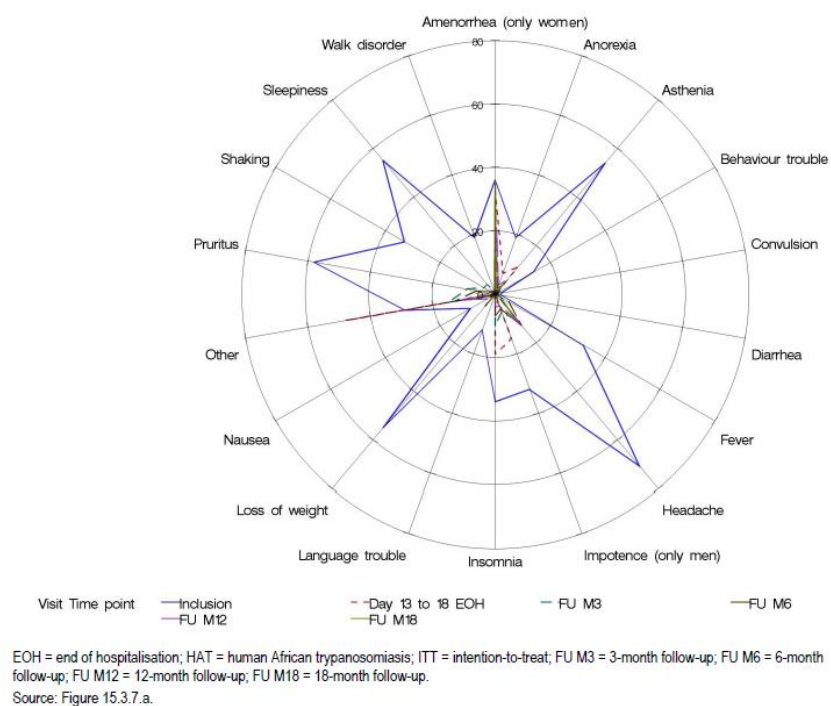


Figure 5. Fexinidazole Group : Radar Charts. Presence of HAT Clinical Signs and Symptoms over Time. ITT Population.

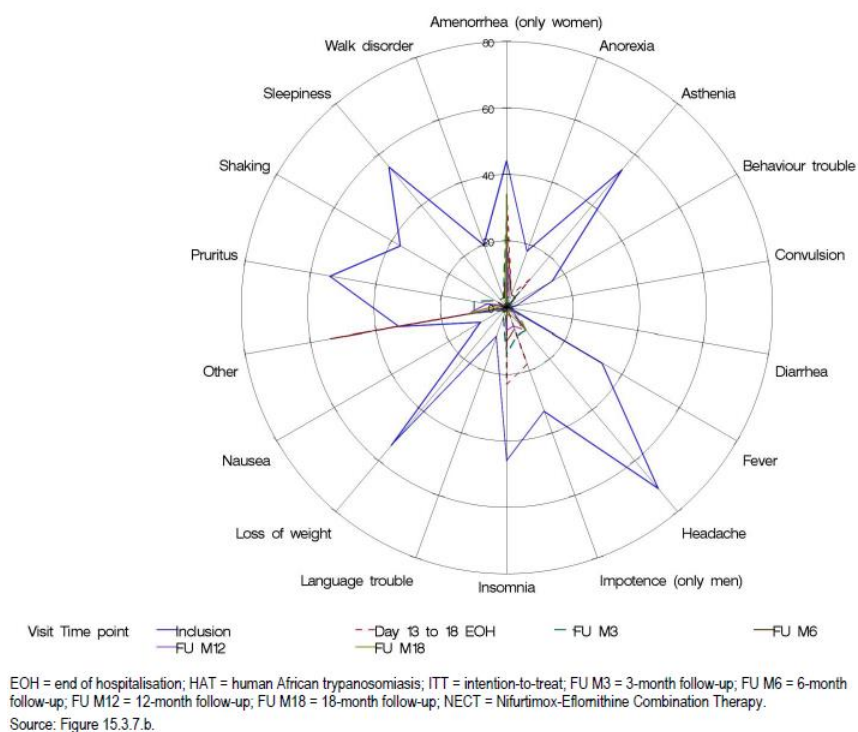


Figure 6. NECT Group : Radar Charts. Presence of HAT Clinical Signs and Symptoms over Time. ITT Population.

The blue solid line represents array of signs and symptoms at baseline, whereas the contracted red-dotted lines are from end of hospitalization. This rapid improvement in signs and symptoms is consistent with the time courses of reduction of parasites and CSF WBC.

The Karnofsky Performance Scale Index provides an assessment of functional impairment.⁸ The assessments were conducted in study FEX004 at baseline and specified visits post dosing. Patients with a Karnofsky score < 60 were excluded in the study.

Figure 7 depicts the evolution of the Karnofsky over time for the ITT population of FEX004. From the median plots ($\pm$ interquartile range), a sustainable improvement in Karnofsky score is seen for both FEXI and NECT groups over time. For failures, there is an ultimate decrement median score at failure. This is especially apparent for those patients who were CSF trypanosome positive at failure. The Karnofsky score may be a clinically useful index for tracking patients over time in clinical practice.⁹

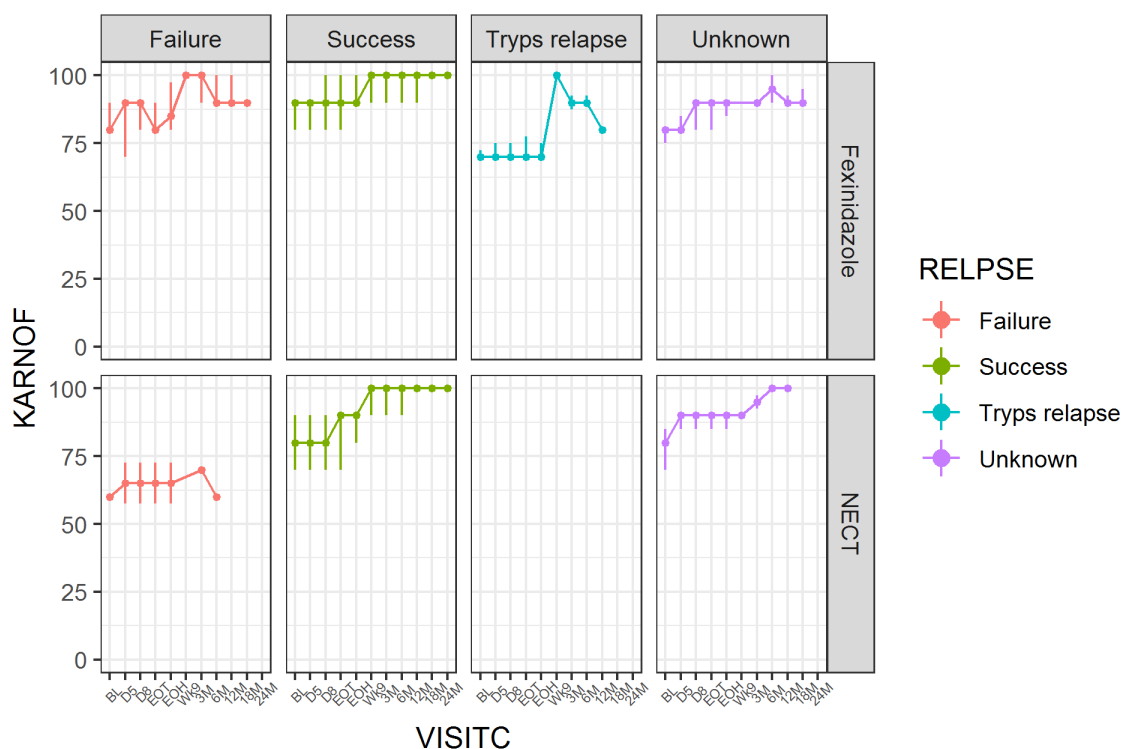


Figure 7. Median ($\pm$ Interquartile Range) Karnofsky Performance Scale Index over time (ITT population). Data on file.

Therefore, looking at parameters of initial response to therapy (CSF trypanosomes, CSF WBC, clinical signs and symptoms, and Karnofsky Performance Scale Index) both Fexi and NECT are highly efficacious and undifferentiated.

⁸ KARNOFSKY DA, BURCHENAL JH, et al. Experimental observations on the effects of the nitrogen mustards on neoplastic tissues. Cancer Res. 1947 Jan;7(1):50.

⁹ http://npcrc.org/files/news/karnofsky_performance_scale.pdf

5. Post-hoc analyses of severity as a covariate of a successful outcome.

During the review of fexinidazole by CHMP, there was a major clinical objection that patients with CNS involvement have lesser efficacy, even though the cure rate at 18 months was within the previously agreed to non-inferiority margin. This observation was made without prospectively defined statistical analyses or sample size. A subsequent request by CHMP was to better describe the efficacy in advanced stage 2 patients based on raw data analysis of subjects with and without trypanosomes in the CSF.

It had been noted that subjects with a baseline CSF WBC level > 100 cells/ μ L were more at risk for relapse per the protocol definition and, if that were the case, then treatment should be steered toward NECT rather than FEXI. A clinical scientific advisory group was convened at the request of the EMA to determine if a clinical score for ambulatory use could adequately predict the probability of drug failure. If the ambulatory clinical score was sufficiently robust, then its use would obviate the need for lumbar puncture for staging for FEXI vs NECT as well. The outcome of this meeting held on 25 September 2018 was that there was no agreement on the preliminary clinical score that had been developed.¹⁰

Since this meeting several graphs have been created to better understand and visualize this point.

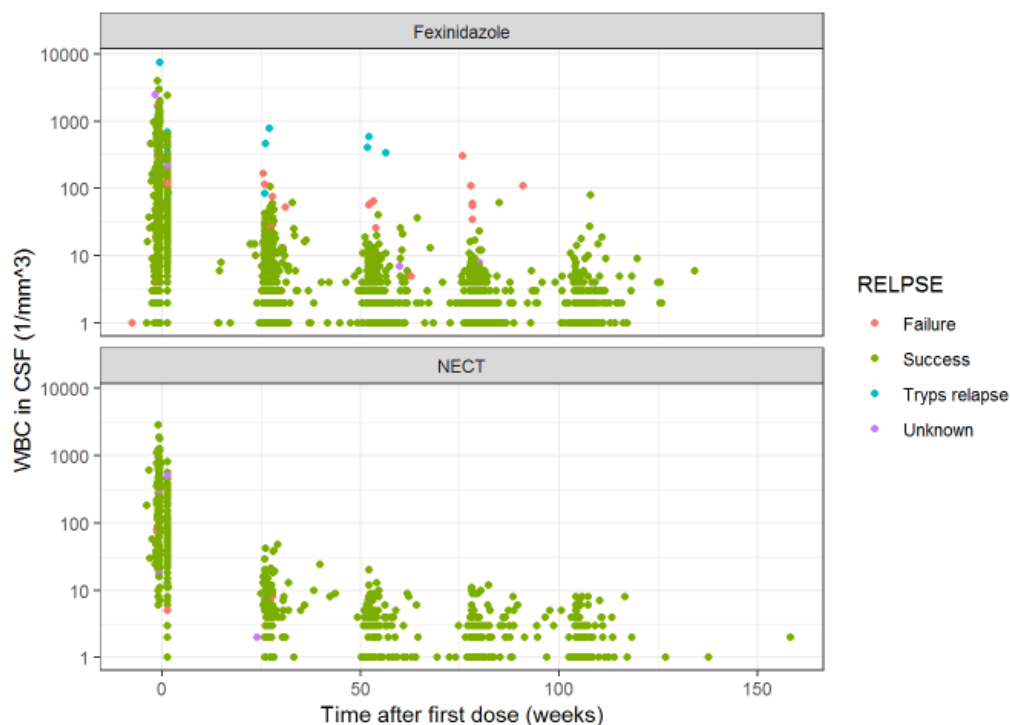


Figure 8. Dot-plot of WBC in CSF vs Time post First Dose for FEXI and NECT. Each dot represents one patient. DNDiFEX004 Clinical Study Report. Data on file.

From the graph above one can see that there is large variability in baseline CSF WBCs. Although the failures generally have a baseline count > 100, it is not uniformly so. There were 3 patients (Subjects 02-023, 02-040 and 07-008) with a CSF WBC count ≤ 100 who relapsed, and there were 218 patients with a

¹⁰ https://www.ema.europa.eu/en/documents/medicine-outside-eu/fexinidazole-winthrop-assessment-report_en-0.pdf Update 4/3/2019

CSF WBC count > 100 that met the protocol definition for success. At baseline 153/394 subjects (39%) had a CSF WBC $\leq$ 100.

Over the weeks of follow-up there was a general decline in WBCs. Those who failed remained at the upper part of dot plots for the respective post-treatment visits. Those failures who were CSF trypanosome positive at failure had CSF WBCs that were for the most part > ~100 post-treatment.

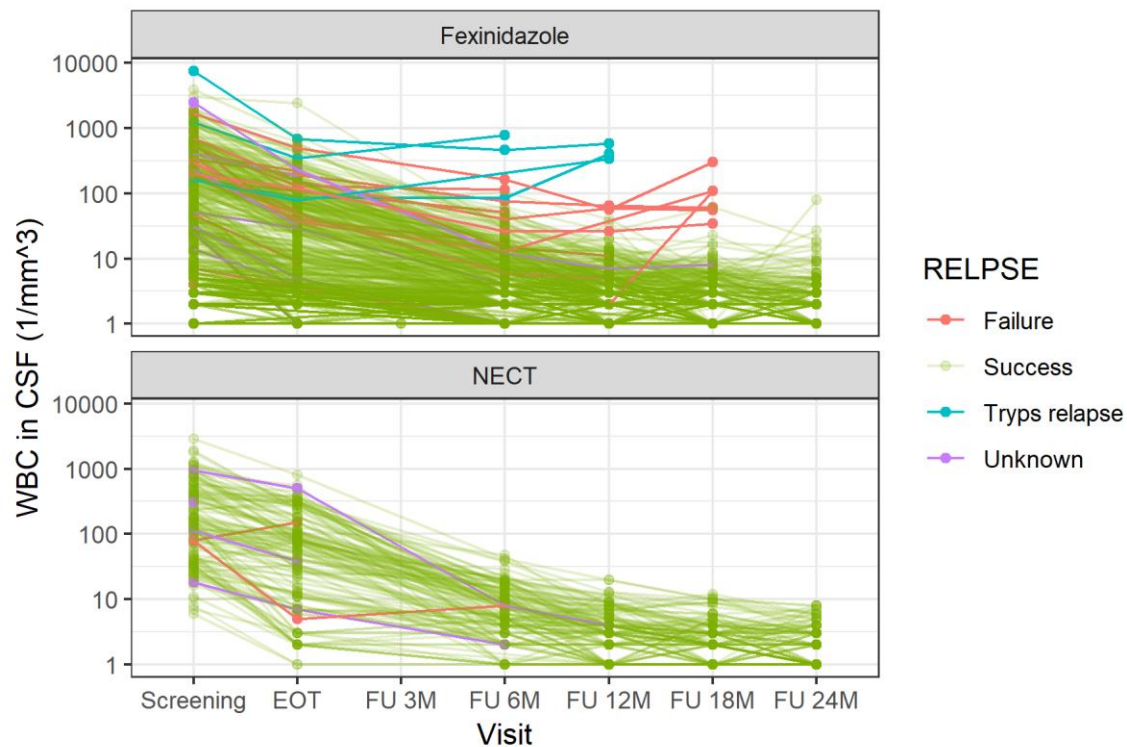


Figure 9. Spaghetti-plot of WBC in CSF vs Time post First Dose for FEXI and NECT. Each dot represents one patient. DNDiFEX004 Clinical Study Report. Data on file.

In a separate post-hoc analysis of pooled data from FEX-004, -005, -006,¹¹ the success rate predicted for patients with CSF WBC < 100 cells/ μ L is > 99%. The relative risk of failure with WBC > 100 was 16.1 and associated with a sensitivity of 87.5% and a specificity of 72.1% with a low positive predictive value (PPV=11%; correct prediction of failure).

One approach to using the CSF WBC > 100 cutoff is that lumbar punctures be performed in all subjects prior to treatment selection with the goal of minimizing treatment failures. There is no mention that this recommendation also took into consideration the additional cost, time and morbidity associated with conducting a LP on all patients. Notably, headache can occur in up to 25-60% of patients who undergo LP, and backache can occur in up to 40% of patients. More serious adverse events are rare and include cerebral herniation, intracranial subdural hemorrhage, spinal epidural hemorrhage, and

¹¹ "Prediction of success of fexinidazole in the treatment of HAT and prediction of the progress of disease with lumbar puncture". Draft report by Bruno Scherrer, PhD, MBA (Independent Consultant in Biostatistics) in collaboration with Nelly Braquet, (senior statistician at Venn Life Paris/study statistician FEX004).

infection. Even with a relatively sensitive biomarker (test for severe disease), the low rate of treatment failure implies that a relatively small fraction of the patients classified as being at high risk of relapse will actually relapse within 18 months. In particular, the PPV of 11% implies that when a failure is predicted because the CSF WBC is > 100 cells/ μ L, then 89% will have unnecessarily received an LP and subsequent NECT.

Another approach would be to avoid the LP by using a sentinel warning symptom scoring mechanism with similar sensitivity, specificity and positive predictive value to the *CSF WBC* > 100 cells/ μ L test. While this is possible, one is still left with the probability of treating unnecessarily with NECT a greater proportion of individuals who meet the test criteria than is truly merited, especially given the large number of patients with a CSF WBC > 100 at baseline who were successfully treated with fexinidazole.

A third approach would be to treat all patients with fexinidazole without conducting an LP or a symptom score. There are several advantages to this approach including cost (related to supply chain with NECT vs. FEXI) and administratively (to give an IV drug vs. oral, especially for patients in remote settings) and the cost, time and morbidity associated with an LP. In addition, patients may not seek treatment for fear of an LP. The potential risk with this approach is that some patients will need to be retreated with NECT within 18 months. While retreatment is not ideal, it would only need to be done in a very small percentage of patients who would likely have a slow onset of signs and symptoms who could therefore be readily identified and treated as opposed to a rapid and severe relapse.

Recall that failure or success is defined temporally by outcome at month 18 post-treatment. Fexinidazole was uniformly successful at end of treatment, meeting the goals of clinical and parasitological success. The failures that did eventually occur were spread out over time indicating that the treatment was not durable in some cases. Based on NECT historical Kaplan-Meier cumulative failure curves over 24 months, and the fact that the FEXI curve was superimposable (see Figure 1 and Table 2), it would be reasonable to treat all patients with Stage 2 HAT without LP staging. In addition, according to the advice of the CHMP for patients “*whose central nervous system is already severely affected*”, fexinidazole should only be given under strict supervision in hospital when no other adequate treatment is available or tolerated.”¹² The diagnostic criteria for these patients, diagnostic procedures performed, and choice of treatment should be a matter of informed clinical judgement in the hands of the clinician.

A final point related to severity of disease is that most HAT studies (including FEX004) exclude patients with a Karnofsky score < 60 . As there is no experience with FEXI in this more severe population, it would be logical to treat them with NECT over FEXI until additional data are generated from subsequent clinical studies or the post-approval pharmacovigilance activities as part of the Risk Management Plan for fexinidazole.

¹² <https://www.ema.europa.eu/en/news/chmp-recommends-first-oral-only-treatment-sleeping-sickness>. Accessed 29 November 2018.

6. Adherence

Compliance with treatment regimen in the pivotal study FEX004 was determined to be high for both tablet ingestion and food intake. The real-world effectiveness is being studied in the ongoing Phase 3b study FEX009.¹³

A Phase 4 pharmacovigilance study is planned once the FEX009 study has completed. Assuming that treatment outcome is a function of pharmacokinetic exposure to pharmacologically active parent and 2 metabolites, adherence to treatment for both tablets and food intake will be important. Techniques to assess adherence in low- to middle-income country field studies have been developed and would be recommended for a Phase 4 pharmacovigilance study.

Summary

Fexinidazole is statistically non-inferior to NECT in curing late Stage 2 Human African Trypanosomiasis at 18 months post-treatment as measured by the proportion of patients with no evidence of trypanosomes in any body fluid; not requiring rescue medication; and, **cerebrospinal fluid white blood cell count ≤ 20 cells/ μ L.**

Patients receiving fexinidazole and NECT achieved goals of parasitological and clinical success by end of treatment (Day 11) as evidenced by marked clinical and neuropsychiatric improvement, **no evidence of trypanosomes in any body fluid and a reduction in CSF WBC.**

Cumulative probability of success curves indicate that fexinidazole is indistinguishable from historical NECT in success over time.

For patients who relapsed after receiving fexinidazole, the time course to relapse was slow, with the small number of relapses occurring over months to years. The clinical course of relapse in general was indolent and mild, not inconsistent with the historical rate of progression of signs and symptoms for primary infections.

Individual patients treated with fexinidazole who relapsed in the months following treatment for the most part had a CSF WBC > 100 (median 147 cells/ μ L) at screening. For patients with CSF WBC < 100 , the success rate was $> 99\%$. However, many patients ($N=167$) in the combined fexinidazole data base with WBCs > 100 were successfully treated and considered cured at 18 months. The utility of CSF WBC > 100 in prospective clinical treatment decisions is open to debate as is the need for any CSF measurement or any cutpoint.

A clinical scoring system using 5 common warning symptoms referable to **Stage 2 HAT has been developed and could provide an alternative clinical test in lieu of lumbar puncture in those patients considered to have more severe CNS involvement.** With a score ≥ 8 , it could provide a non-invasive test to clinically discern those patients with more severe central nervous system involvement who might be at greater risk of **failing treatment with fexinidazole.** The post-hoc analyses used to develop and verify the clinical severity score have not been prospectively confirmed. While this approach would eliminate the requirement for an LP, a clinical score **surrogate instead of an LP-derived CSF WBC >100 would lead**

¹³ DNDi. An Open-label Study Assessing Effectiveness, Safety and Compliance With Fexinidazole in Patients With Human African Trypanosomiasis Due to *T.b. Gambiense* at Any Stage. ClinicalTrials.gov Identifier: NCT03025789

to 87% of patients being treated with NECT when an oral regimen of FEXI would have been curative at 18 months.

In conclusion, fexinidazole is a novel oral medication demonstrated to have a highly favorable risk / benefit profile in both Stage 1 and Stage 2 HAT. It rapidly treats the symptomatic and objective abnormalities associated with HAT and, in the rare instances of relapse, the clinical course is sufficiently indolent, allowing time to seek subsequent medical attention. It represents an important breakthrough in the treatment armamentarium for this rare yet serious disease. NECT remains an effective treatment for HAT and should be made available when logistically possible for FEXI patients who have relapsed; patients not previously studied in FEXI clinical trials (eg patients with a Karnofsky score < 60); or patients who can't tolerate or have unacceptable side effects to FEXI; or anyone deemed appropriate by a properly informed health professional.

Recommendations:

- Fexinidazole should be first-line therapy for Stage 1 and Stage 2 HAT.
- Lumbar puncture should neither be required nor proscribed in the case of uncertainty about diagnosis or severity of disease.
- For patients with more severe CNS manifestations at time of consultation, consider the probability of durable cure for patients whose follow up may be uncertain.
- Patients with a Karnofsky score <60 should be treated with NECT until further data are generated with fexinidazole in this population
- Adherence to treatment and food should be encouraged and monitored in the Phase 4 pharmacovigilance study.
- Inform all patients with late Stage 2 HAT, that should signs and symptoms recur, to seek qualified medical consultation.

Disclaimer: This draft report represents a living document/concept sheet submitted to further dialog in the global health community. Further analyses of the various studies conducted to date may provide additional insights into treatment options and the disease in general.

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